Assignment 5: Semantics II - Welcome to the World of Mario

Ling 201 – Fall 2026

(100 pts total)

Please **do not use external sources such as the internet or AI to find the answers**. If you need an **extension**, please ask before the deadline.

Part 1. Presupposition Test *[difficulty level: ☆☆☆☆]*

For each pair of sentences below, determine **whether or not sentence (a) presupposes sentence (b)**, and **justify your answer using semantic reasoning** (truth conditions, etc.). Do not panic, if you follow the steps shown in the **example**, you will find the questions manageable.

(6 points/item, 30 points total)

Example: (a). Mario ate the cookie.

(b). There exists a cookie.

As we said for (a) to presuppose (b), (a) must entail (b) first, since **a presupposition relation is a special kind of entailment relation**.

> **step 1**: (a) **ENTAILS** (b), if Mario ate the cookie, then there must be at least one cookie for him to eat. Hence, if (a) is **TRUE**, (b) must also be **TRUE**.

We have also mentioned in class, that presupposition survives negation, which means that **the negation of (a) must entail (b)** as well to form a presupposition relation.

> **step 2**: **NOT-(a) ENTAILS** (b), if Mario *DID NOT* eat the cookie, then the cookie must still exist. Hence, if NOT-(a) is **TRUE**, (b) must also be **TRUE**.

Two steps **together**: if (a) is **TRUE**, (b) must also be **TRUE**;

if NOT- (a) is **TRUE**, (b) must also be **TRUE**.

This means, sentence (a) **PRESUPPOSES** sentence (b).

Note: Please explicitly state whether each entailment holds ((a) entails (b), NOT-(a) entails (b)) before giving your final answer.

1. (a). Toad texted Toadette again.

   (b). Toad texted Toadette before.

2. (a). Baby Luigi did his homework.

   (b). Baby Luigi redid his homework.

3. (a). The world of Mario is fantastic.

   (b). Bowser knows that the world of Mario is fantastic.

4. (a). What Boo wanted was the crown.

   (b). Boo wanted something.

5. (a). Daisy finished reading the paper.

   (b). Daisy started reading the paper.

Part 2. Set Theory ***[difficulty level: ☆☆]***

Here's a photo of characters in the game *Mario Kart*.



Answer the following questions, assuming the following subsets:

- M = { x | x has a giant mouth } = { Yoshi, Birdo, Wario, Bowser, Petey Piranha, King Boo, Waluigi }
- N = { x | x has a big nose } = { Mario, Yoshi, Baby Mario, Luigi, Baby Luigi, Wario, Bowser, Waluigi }
- H = { x | x is humanoid } = { Yoshi, Toad, Birdo, Toadette, Bowser, Petey Piranha, King Boo }
- C = { x | x wears a cap } = { Mario, Baby Mario, Luigi, Baby Luigi, Wario, Waluigi }
- G = { x | x 'wears' green } = { Peach, Yoshi, Luigi, Daisy, Baby Luigi, Bowser, Petey Piranha} *(Here, Peach and Daisy are included because of their earrings.)*
- T = { x | x's tongue is visible} = { Mario, Yoshi, Bowser, Petey Piranha, King Boo }

- $K = \{x \mid x \text{ is crowned}\}$ = { Peach, Daisy, King Boo }

1. **State whether the expression is true or false** and **briefly explain why**. *(4 points/item, 24 points total)*

   a. $|M| = |H|$

   b. $G \cap T \neq \emptyset$

   c. King Boo $\in$ ($N \cap H$)

   d. $T \subseteq M$

   e. Peach $\in C'$

   f. $K - M$ = { King Boo }

2. **Calculate** the following and **describe the resulting set in words**. (*4 points/item, 24 points total)*

   Example: $G - K$

   calculated set: *{Yoshi, Luigi, Baby Luigi, Bowser, Petey Piranha}*

   description in words: *characters who wear green but are not crowned*

   a. $N \cap H$

   b. $G - C$

c. $K \cap H'$

d. $(K \cup N)'$

e. $T - (K \cup M)$

f. $(H - M) \cup (K \cap T)$

3. Translate the following sentences into set-theoretic notation by following semantics of quantifiers. (*4 points/item, 8 points total*)

   a. Some characters with giant mouths have big noses.

b. No crowned characters wear a cap.

4. Define two additional subsets of your own to add to the top list. Use **both predicate notation and list notation for each set**. *(3 points/item, 6 points total)*

   i. ____ = { x | x_____} = { _____}

   ii. ____ = {x | x_____} = { _____}

5. Create two phrases using the sets you defined in Question 4 and the sets provided above. Across your two answers, you must use each of your newly defined sets at least once. Represent the phrase in **set notation**, then calculate the resulting set and **write it in list notation**. You must use **at least two operators** in each expression, out of the following: union (∪), intersection (∩), difference (–), complement (′).

   (4 points/item, 8 points total)

   Example phrase: *Humanoid characters who have big noses or wear caps*

   in set notation: $H \cap (N \cup C)$ = { Yoshi, Bowser }

   a. phrase 1: _____

      in set notation: _____= { _____ }

   b. phrase 2: _____

      in set notation: _____= { _____ }